



Lokmanya Tilak Jankalyan Shikshan Sanstha's
PRIYADARSHINI COLLEGE OF ENGINEERING, NAGPUR
(An Autonomous Institute affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)



DEPARTMENT OF INFORMATION TECHNOLOGY

CYBER SECURITY IN BUSINESS - LAB MANUAL

Subject code: 24UHBIT6P

Semester: VI



DEPARTMENT OF INFORMATION TECHNOLOGY

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LOKMANYA TILAK JANKALYAN SHIKSHAN SANSTHA'S

PRIYADARSHINI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to R.T.M. Nagpur University, Nagpur

Accredited with Grade 'A+' by NAAC

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UNIVERSITY CODE No. : 278



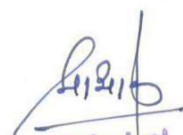
Vision

To become one of the India's leading Engineering Institutes in both education and research. We are committed to provide quality and state-of-the-art technical education to our students so that they become Technologically competent and in turn contribute for creating a great society.

Mission

1. Fostering a dynamic learning environment that equips students with Technical expertise, problem-solving skills and a deep commitment to ethical practices.
2. To cultivate a culture of innovation, incubation, research and entrepreneurship that drives technological advancements.
3. To uphold the spirit of mutual excellence while interacting with stake holders of our Institutional ecosystem.
4. Promoting lifelong learning, professional growth and ensuring holistic development of students and the well being of society.




Principal
Priyadarshini College of Engg.
Nagpur.

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Department of Information Technology

Date: 16/09/2023

Vision

To nurture the students with latest technology and research, helping them to become valuable contributors who can address the changing needs of the IT industry and society.

Mission

1. To create a holistic learning environment which empower the emerging graduates with advanced technical and value based education.
2. To attain professional excellence in the field of Information Technology and related areas.
3. To encourage transformative learning for research addressing the changing needs of society and fostering lifelong learning.

Bansod
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Priyadarshini College of Engg
Nagpur-19

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PROGRAM EDUCATIONAL OBJECTIVES :

PEOs	Program Educational Objectives Statements
PEO1	Demonstrate strong technical expertise and uphold ethical values to excel in their professional careers.
PEO2	Apply advanced skills and knowledge in Information Technology to address real-world challenges and contribute to industry success.
PEO3	Pursue lifelong learning and engage in research and innovation to address societal needs and advance the field of Information Technology.



DEPARTMENT OF INFORMATION TECHNOLOGY

PROGRAM OUTCOMES:

Engineering Graduates will be able to:

POs	Program Outcomes
PO1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.(WK8).
PO5	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO6	The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO8	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
PO10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)



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DEPARTMENT OF INFORMATION TECHNOLOGY

PROGRAM SPECIFIC OUTCOMES :

PSOs	Program Specific Outcomes
PSO1	An ability to apply mathematical foundations, algorithmic principles and computer science theory in the modeling and design of software systems of varying complexity.
PSO2	An ability to work with Open-Source Software and use off the shelf utilities for program integration.



DEPARTMENT OF INFORMATION TECHNOLOGY

COURSE OUTCOMES :

Course Outcomes	Statement
CO1	Identify and analyze cyber security threats affecting business systems, networks, and digital assets through hands-on practical experiments.
CO2	Implement basic security mechanisms such as firewalls, VPNs, antivirus solutions, and access controls to protect business environments.
CO3	Apply ethical hacking and security testing techniques (password strength analysis, phishing detection, malware simulation) in a controlled and legal environment.

CO-PO & PSO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	3	3	1	1	1	1	1	1	1	3	2
CO2	3	2	3	1	3	2	1	1	1	1	1	1	3	3
CO3	2	3	2	3	3	2	1	2	1	2	1	1	2	3



DEPARTMENT OF INFORMATION TECHNOLOGY

List of Practical's:

Sr. No.	Name of Practical
1	Setting up user accounts with strong passwords and role based access control.
2	Implementing multi factor authentication(MFA).
3	Conducting password strength analysis and controlled cracking simulation
4	Configuring a basic firewall on a system or virtual network.
5	Setting up a VPN and secure remote connection between system.
6	Implementing IDS/IPS for detecting network intrusions.
7	Analyzing network traffic using wireshark to detect vulnerabilities.
8	Simulating malware attacks in a controlled virtual environment.
9	Phishing attack simulation and detection exercises.
10	Implementing antivirus/ endpoint protection and malware removal.
11	Case study analysis of legal and ethical cyber security incidents.
Content Beyond Syllabus	
12	To study advanced cyber security concepts, tools, and real-world business security practices beyond the prescribed syllabus, enabling students to gain industry-oriented knowledge and practical awareness.



PRACTICAL NO.1

Aim: Setting up user accounts with strong passwords and role based access control.

Theory:

In modern business organizations, information systems are accessed by multiple users such as employees, managers, and administrators. Without proper access control mechanisms, organizations are exposed to risks including data breaches, insider threats, and unauthorized access. This experiment demonstrates how strong passwords and RBAC enhance cyber security in business environments.

User Account Management:

User account management ensures that each user has a unique identity, enabling accountability and monitoring.

Strong Password Policy:

Strong passwords prevent brute-force and dictionary attacks by enforcing complexity, length, and expiration policies.

Role-Based Access Control (RBAC):

RBAC assigns permissions based on roles rather than individuals, reducing administrative complexity and security risks.

Principle of Least Privilege:

Users are granted only the permissions required to perform their tasks, minimizing potential attack surfaces. This approach is widely adopted in enterprise cyber security frameworks and compliance standards.

User Account Management:

User account management ensures that each user has a unique identity, enabling accountability and monitoring.

Strong Password Policy:

Strong passwords prevent brute-force and dictionary attacks by enforcing complexity, length, and expiration policies.

Procedure:

1. Log in with administrator privileges.
2. Create multiple user accounts.
3. Configure password policies.
4. Define user roles.
5. Assign permissions to roles.
6. Map users to roles.
7. Test access controls.

Output

- Strong password policies enforced.
- RBAC implemented successfully.
- Unauthorized access attempts restricted.



Result:

The experiment successfully demonstrated how strong password policies and role-based access control enhance business cyber security by ensuring secure and controlled system access.

Viva Questions and Answers

Q1. What is RBAC?

A. Role-Based Access Control.

Q2. Why are strong passwords important?

A. They protect against unauthorized access.

Q3. What is least privilege?

A. Granting minimal permissions.

Q4. Authentication vs Authorization?

A. Authentication verifies identity; authorization grants access.

Q5. Name two password attacks.

A. Brute-force and dictionary attacks.



PRACTICAL NO: 2

Aim: Implementing multi factor authentication(MFA).

Theory:

In modern business environments, cyber attacks such as phishing, credential theft, and brute-force attacks are increasingly common. Traditional password-based authentication alone is insufficient to protect sensitive business data. Multi-Factor Authentication (MFA) provides an additional layer of security by requiring users to verify their identity using multiple factors.

Authentication is the process of verifying the identity of a user accessing a system. MFA strengthens authentication by combining two or more verification factors.

Authentication Factors:

1. Something you know – passwords or PINs
2. Something you have – OTP, mobile devices, hardware tokens
3. Something you are – biometrics such as fingerprints or facial recognition

MFA is widely used in business applications such as banking, cloud services, and enterprise systems. It significantly reduces the risk of unauthorized access even if passwords are compromised.

MFA also supports regulatory compliance and improves overall cyber security posture. Authentication is the process of verifying the identity of a user accessing a system. MFA strengthens authentication by combining two or more verification factors.

Procedure:

1. Log in with administrator privileges.
2. Navigate to security settings and enable MFA.
3. Select secondary authentication methods such as OTP or authenticator apps.
4. Register user devices for MFA.
5. Enroll users for MFA during first login.
6. Test MFA by performing login attempts.

Output:

- MFA enabled successfully
- Users required secondary authentication
- Unauthorized access attempts blocked

Result:

This experiment demonstrated the implementation of Multi-Factor Authentication in a business environment. MFA significantly enhances security by preventing unauthorized access even when login credentials are compromised.

Viva Questions and Answers

Q1. What is MFA?

A. Multi-Factor Authentication requires multiple verification methods.

Q2. Why is MFA important?

A. It improves security and prevents unauthorized access.



Q3. Name three authentication factors.

A. Knowledge, possession, and biometric factors.

Q4. What is OTP?

A. One-Time Password.

Q5. How does MFA prevent phishing?

A. Password theft alone cannot grant access.

Q6. Where is MFA commonly used?

A. Banking, cloud services, enterprise systems.



PRACTICAL NO. 3

Aim: Conducting password strength analysis and controlled cracking simulation.

Theory:

Passwords are the first line of defense in business information systems. Weak passwords expose organizations to cyber-attacks such as data breaches, financial fraud, and identity theft.

Password strength depends on length, character variety, unpredictability, and entropy. Password cracking is the process of recovering passwords from stored hashes using techniques like dictionary and brute-force attacks. This experiment demonstrates these concepts ethically.

Procedure:

Part A: Password Strength Analysis:

Password Strength Analysis is the process of evaluating a password's resistance to guessing or cracking attacks. It determines whether a password can protect systems from unauthorized access.

1. Create a sample password list.
2. Analyze length, character types, and entropy.
3. Classify passwords as weak, moderate, or strong.

Part B: Controlled Cracking Simulation:

Controlled Cracking Simulation is an ethical and authorized experiment conducted in a lab environment using test passwords and systems. Its purpose is educational, not malicious.

4. Convert passwords into hash format.
5. Perform dictionary and rule-based attacks.
6. Observe and record results.

Result: The experiment proves that weak and predictable passwords are easily cracked, while long and complex passwords provide stronger protection. Hence, the aim of the experiment is successfully achieved.



Viva Questions and Answers:

1. What is password strength analysis?

Answer: Password strength analysis evaluates how resistant a password is to guessing or cracking attacks based on length, complexity, and unpredictability.

2. Why is password security important in business?

Answer: Weak passwords can lead to unauthorized access, data breaches, financial loss, and damage to business reputation.

3. What factors determine password strength?

Answer: Password length, use of uppercase and lowercase letters, numbers, special characters, and randomness.



PRACTICAL NO. 4

Aim: Configuring a basic firewall on a system or virtual network.

Theory:

Firewall: A firewall is a network security mechanism that monitors and controls incoming and outgoing network traffic based on predefined security rules. It acts as a barrier between a trusted internal network and an untrusted external network such as the Internet.

In **business environments**, firewalls are used to:

- Protect sensitive company data
- Prevent unauthorized access
- Reduce the risk of malware and cyberattacks
- Ensure secure communication between systems

Types of Firewalls

1. **Software Firewall** – Installed on individual systems
2. **Hardware Firewall** – Dedicated physical device
3. **Network Firewall** – Protects entire networks
4. **Stateful Firewall** – Tracks connection states

Procedure:(Using Linux – UFW Firewall)

- Open the terminal
- Update system packages
- Enable the firewall
- Allow required services (SSH, HTTP, HTTPS)
- Block unused or insecure ports
- Check firewall status

Sample Commands / Code:

```
sudo apt update
```

```
sudo apt install ufw
```

```
sudo ufw reset
```

```
sudo ufw default deny incoming
```

```
sudo ufw default allow outgoing
```

```
sudo ufw allow ssh
```




```
sudo ufw allow http
```

```
sudo ufw allow https
```

```
sudo ufw enable
```

```
sudo ufw status verbose
```

Observations / Output:

- Firewall status: **Active**
- Unauthorized ports: **Blocked**
- Authorized services: **Allowed**
- System protected from unwanted network access

Result:

The basic firewall was successfully configured on the system. Incoming unauthorized network traffic was blocked while allowing necessary business services. Hence, the aim of the experiment was achieved successfully.

Viva Questions and Answers

Q1. What is a firewall?

Answer: A firewall is a security system that monitors and controls network traffic based on predefined rules.

Q2. Why is a firewall important in business organizations?

Answer: It protects sensitive business data, prevents cyberattacks, and controls unauthorized access.

Q3. What is UFW?

Answer: UFW (Uncomplicated Firewall) is a user-friendly firewall management tool in Linux.

Q4. What is port blocking?

Answer: Blocking unused or insecure network ports to prevent attacks.

Q5. Difference between software and hardware firewall?

Answer: Software firewall runs on a system, while hardware firewall is a dedicated physical device.



PRACTICAL NO: 5

Aim: Setting up a VPN and secure remote connection between system.

Theory:

A Virtual Private Network (VPN) is a secure communication technology that creates an **encrypted tunnel** between two systems over an untrusted network such as the Internet. It ensures that data transmitted between systems remains private and protected from interception.

Need for VPN in Business:

In modern business environments:

- Employees work remotely
- Branch offices are geographically distributed
- Sensitive data is transmitted over public networks

VPNs help businesses by:

- Securing remote employee access
- Protecting confidential business information
- Preventing data theft and eavesdropping
- Ensuring compliance with security policies

Procedure: (Using OpenVPN – Conceptual Procedure)

- Install VPN software on server and client systems
- Configure VPN server with encryption and authentication
- Create VPN client configuration
- Start VPN service on server
- Connect client to VPN server
- Verify secure remote access

Sample Commands / Code (Linux):

```
sudo apt update
```

```
sudo apt install openvpn
```

```
sudo openvpn --config client.ovpn
```

Observations / Output:

- VPN tunnel established successful



- Remote system securely connected
- Data transmission encrypted
- Internal network resources accessible remotely

Real-World Business Applications:

- IT companies enable secure work-from-home access
- Banks use VPNs for secure branch communication
- Corporate employees access internal servers remotely
- Multinational companies connect global offices securely

Result:

The VPN and secure remote connection between systems were successfully established. The data transmitted between systems was encrypted, ensuring confidentiality and security. Hence, the **aim of** the experiment was achieved successfully.

Viva Questions and Answers:

Q1. What is a VPN?

Answer: A VPN is a technology that creates a secure, encrypted connection over a public network.

Q2. Why is VPN important for business organizations?

Answer: It protects business data during remote access and prevents cyber threats.

Q3. What is encryption in VPN?

Answer: Encryption converts data into unreadable format to prevent unauthorized access.

Q4. What is tunneling in VPN?

Answer: Tunneling is the process of securely encapsulating data packets.

Q5. Difference between VPN and proxy?

Answer: VPN encrypts all traffic, while proxy only forwards traffic without full encryption.

Q6. What is remote access VPN?

Answer: A VPN that allows employees to connect securely to company networks from remote locations.



PRACTICAL NO: 6

Aim: Implementing IDS/IPS for detecting network intrusions.

Theory:

Intrusion Detection System (IDS):

An **IDS** monitors network traffic and system activities to detect suspicious behavior or policy violations. It generates alerts when potential intrusions are detected.

Intrusion Prevention System (IPS):

An **IPS** is an advanced form of IDS that not only detects intrusions but also **actively blocks or prevents** malicious traffic in real time.

Importance of IDS/IPS in Business:

Businesses face threats such as:

- Unauthorized access
- Malware attacks
- Data breaches
- Denial-of-Service (DoS) attacks

IDS/IPS helps businesses by:

- Monitoring network traffic continuously
- Detecting cyberattacks early
- Preventing system compromise
- Protecting sensitive business data

Types of IDS/IPS:

- **Network-based IDS/IPS (NIDS/NIPS)**
- **Host-based IDS/IPS (HIDS/HIPS)**
- **Signature-based detection**
- **Anomaly-based detection**

Tools / Software Required:

- Operating System: Linux (Ubuntu / Kali)
- IDS/IPS Tool: **Snort / Suricata**
- Network interface
- Administrator/root privileges
- Virtual Machine (optional)

Experimental Setup:



- One system acting as IDS/IPS
- Network traffic source (test traffic)
- IDS/IPS rules configured

Procedure:

(Using Snort – Conceptual Procedure)

- Install IDS/IPS tool
- Configure network interface
- Load default rule sets
- Start IDS in detection modeSimulate suspicious traffic
- Observe alerts and logs
- Enable prevention mode (IPS)

Sample Commands / Code:

```
sudo apt update
sudo apt install snort
sudo snort -T -c /etc/snort/snort.conf
sudo snort -A console -q -c /etc/snort/snort.conf -i eth0
```

Observations / Output:

- Alerts generated for suspicious activities
- Malicious packets detected
- Intrusion attempts logged
- Network traffic monitored in real time

Real-World Business Applications:

- Banks monitor transactions to detect cyber fraud
- E-commerce companies detect hacking attempts
- Corporate networks prevent unauthorized access
- SOC (Security Operations Center) uses IDS/IPS for monitoring

Result:

The IDS/IPS was successfully implemented and detected network intrusions. The system generated alerts and prevented unauthorized network activities. Hence, the aim of the experiment was achieved successfully.



Viva Questions and Answers:

Q1. What is IDS?

Answer: IDS is a system that monitors network traffic and alerts when suspicious activity is detected.

Q2. What is IPS?

Answer: IPS detects and automatically blocks malicious network traffic.

Q3. Difference between IDS and IPS?

Answer: IDS only detects attacks, while IPS detects and prevents attacks.

Q4. What is Snort?

Answer: Snort is an open-source IDS/IPS tool used for network intrusion detection.

Q5. What is signature-based detection?

Answer: Detection based on known attack patterns or signatures.

Q6. What are false positives?

Answer: Legitimate traffic incorrectly detected as malicious.



PRACTICAL NO: 7

Aim: Analyzing network traffic using Wireshark to detect vulnerabilities.

Theory:

Network Traffic Analysis:

Network traffic analysis involves monitoring and examining data packets flowing through a network to understand communication patterns, detect anomalies, and identify security threats.

Wireshark: Wireshark is an open-source packet analyzer used for:

- Network troubleshooting
- Security analysis
- Protocol development
- Detection of malicious activities

Wireshark captures packets in real time and displays detailed information such as source IP, destination IP, protocol, and payload.

Importance in Business:

Businesses use network traffic analysis to:

- Detect data leakage
- Identify unauthorized access
- Monitor employee network usage
- Investigate cyber incidents
- Ensure compliance with security policies

Common Vulnerabilities Detected:

- Unencrypted data transmission
- Weak or outdated protocols
- Suspicious IP addresses
- Malware communication
- Man-in-the-Middle attacks

Procedure:

1. Install and launch Wireshark
2. Select the active network interface
3. Start packet capture
4. Generate network traffic (browse websites, ping, etc.)
5. Apply filters to analyze specific traffic
6. Identify suspicious or vulnerable communication
7. Stop capture and save results



Sample Filters / Commands:

```
http
https
tcp.port == 80
tcp.port == 443
ip.addr == 192.168.1.1
dns
```

Observations / Output:

- Packets captured successfully
- Source and destination IP addresses identified
- Protocols analyzed
- Unencrypted and suspicious traffic detected

Result:

Network traffic was successfully captured and analyzed using Wireshark. Vulnerabilities such as unencrypted communication and suspicious network behavior were identified. Hence, the aim of the experiment was achieved successfully.

Viva Questions and Answers

Q1. What is Wireshark?

Answer: Wireshark is an open-source packet capturing and network analysis tool.

Q2. What is packet sniffing?

Answer: Packet sniffing is the process of capturing and analyzing network packets.

Q3. Why is network traffic analysis important in business?

Answer: It helps detect cyber threats, data leakage, and unauthorized access.

Q4. What information does a packet contain?

Answer: Source IP, destination IP, protocol, and payload data.

Q5. What is a display filter in Wireshark?

Answer: A rule used to display specific packets from captured data.



PRACTICAL NO: 8

Aim: Simulating malware attacks in a controlled virtual environment.

Theory:

Malware: Malware (Malicious Software) is any software intentionally designed to damage, disrupt, or gain unauthorized access to systems.

Types of Malware:

Type	Description
Virus	Attaches to files and spreads
Worm	Self-replicates over networks
Trojan	Disguised as legitimate software
Ransomware	Encrypts data for ransom
Spyware	Steals user information

Controlled Virtual Environment:

A **virtual environment** uses Virtual Machines (VMs) to safely simulate attacks without harming real systems.

This is essential for:

- Safe experimentation
- Malware research
- Business security training

Procedure (Safe Simulation Approach):

- Create an isolated virtual machine
- Disable internet access inside VM
- Install antivirus software
- Use simulated malware test files (e.g., EICAR test string)
- Execute the test sample
- Observe system behavior and alerts
- Record detection and response
- Restore VM snapshot

Sample Test Code (EICAR Test – Safe):

X5O!P%#@AP[4PZX54(P^)7CC)7}\$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!\$H+H*

Observations / Output:

- Antivirus alert generated
- Test malware detected successfully



- System resources monitored
- Logs recorded for analysis

Result:

Malware behavior was safely simulated in a controlled virtual environment. The antivirus system successfully detected and responded to the simulated malware. Thus, the aim of the experiment was achieved successfully.

Viva Questions and Answers:

Q1. What is malware?

Answer: Malware is malicious software designed to harm or exploit systems.

Q2. Why use a virtual environment for malware simulation?

Answer: To safely study malware behavior without affecting real systems.

Q3. What is the EICAR test file?

Answer: A harmless test file used to verify antivirus detection.

Q4. Name different types of malware.

Answer: Virus, Worm, Trojan, Ransomware, Spyware.

Q5. Why is malware simulation important in business?

Answer: It helps businesses prepare for real cyberattacks and improve defenses.

Q6. What is ransomware?

Answer: Malware that encrypts data and demands ransom.



PRACTICAL NO: 9

Aim: Phishing attack simulation and detection exercises.

Theory:

Phishing:- Phishing is a social engineering attack where attackers trick users into revealing sensitive information such as: Usernames, Passwords, Credit card details, Business credentials. It is commonly carried out through emails, fake websites, SMS, or social media messages.

Importance in Business:

Phishing is one of the largest causes of business data breaches. Businesses use phishing simulations to:

- Train employees
- Reduce human error
- Improve incident response
- Protect financial and customer data

Detection Techniques:

- Checking sender email domain
- Hovering over links
- Verifying HTTPS and certificates
- Email header analysis
- Using spam filters and antivirus

Observations / Output:

- Phishing characteristics identified
- Fake links detected
- Suspicious email patterns recognized
- Security alerts triggered

Result:

Phishing attack scenarios were successfully simulated and analyzed. Various phishing indicators were detected, and appropriate preventive measures were identified. Hence, the aim of the experiment was achieved successfully.



Viva Questions and Answers:

Q1. What is phishing?

Answer: Phishing is a cyberattack where attackers trick users into revealing sensitive information.

Q2. Why is phishing dangerous for businesses?

Answer: It can lead to data breaches, financial loss, and reputation damage.

Q3. What is spear phishing?

Answer: A targeted phishing attack aimed at specific individuals.



PRACTICAL NO: 10

Aim: Implementing antivirus/ endpoint protection and malware removal.

Theory:

Endpoint Security:- An endpoint is any device connected to a business network such as: Desktops, Laptops, Mobile devices, Servers. Endpoint security protects these devices from cyber threats such as malware, ransomware, spyware, and viruses.

Antivirus Software:

Antivirus software is designed to:

- Detect malicious files
- Prevent malware execution
- Quarantine infected files
- Remove malware safely

3 Endpoint Protection Platform (EPP):

Modern businesses use **Endpoint Protection Platforms (EPP)** which include:

- Antivirus
- Firewall
- Intrusion prevention
- Behavioral analysis
- Centralized management

Importance in Business:

Businesses rely on endpoint protection to:

- Prevent data breaches
- Protect customer and financial data
- Ensure business continuity
- Comply with cybersecurity policies

Procedure:

Step 1: Install Antivirus / Endpoint Protection

- Open antivirus software
- Update virus definitions
- Enable real-time protection

Step 2: Configure Security Settings

- Enable scheduled scans



- Enable automatic quarantine
- Enable firewall (if available)

Step 3: Malware Detection (Safe Simulation)

- Use **EICAR** antivirus test file
- Save the test file on the system
- Observe antivirus response

Result:

Antivirus and endpoint protection were successfully implemented. The system detected and removed simulated malware, ensuring endpoint security. Hence, the aim of the experiment was achieved successfully.

Viva Questions and Answers:

Q1. What is antivirus software?

Answer: Antivirus software detects, prevents, and removes malware from systems.

Q2. What is endpoint security?

Answer: Endpoint security protects devices connected to a network from cyber threats.

Q3. What is malware?

Answer: Malware is malicious software designed to harm or exploit systems.

Q4. What is real-time protection?

Answer: Continuous monitoring of system activity to block threats instantly.



PRACTICAL NO: 11

Aim: Case study analysis of legal and ethical cyber security incidents.

Theory:

Cyber Security Incidents: A cyber security incident is any event that compromises:

Confidentiality, Integrity, Availability of information systems or data .Examples include:Data breaches, Ransomware attacks, Insider threats, Unauthorized data access.

Sample Case Studies (Examples):

Case Study 1: Data Breach in a Financial Company

- Customer financial data exposed
- Legal violation of data protection laws
- Ethical failure in safeguarding customer trust

Case Study 2: Ransomware Attack on a Hospital

- Patient records encrypted
- Legal consequences for service disruption
- Ethical concern regarding patient safety

Case Study 3: Insider Threat in an IT Organization

- Employee misused access privileges
- Breach of confidentiality
- Ethical violation of professional conduct

Real-World Business Relevance:

- Helps organizations comply with cyber laws
- Improves ethical decision-making
- Reduces legal penalties
- Builds customer trust
- Strengthens corporate governance

Result: The selected cyber security incidents were successfully analyzed from legal and ethical perspectives. The study provided insights into business responsibilities, legal compliance, and ethical decision-making. Hence, the aim of the experiment was achieved successfully.

Viva Questions and Answers

Q1. What is a cyber security incident?



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Answer: An event that compromises data or system security.

Q2. Why are legal aspects important in cyber security?

Answer: They ensure compliance, accountability, and protection of user rights.

Q3. What is cyber ethics?

Answer: Moral principles guiding responsible use of technology and data.

Q4. Difference between legal and ethical issues?

Answer: Legal issues violate laws; ethical issues violate moral principles.

Q5. Name a common cyber law in India.

Answer: Information Technology Act, 2000.

Q6. What is a data breach?

Answer: Unauthorized access to sensitive information.



Content beyond Syllabus

PRACTICAL NO: 12

Aim: To study advanced cyber security concepts, tools, and real-world business security practices beyond the prescribed syllabus, enabling students to gain industry-oriented knowledge and practical awareness.

Theory: Cyber Security in Business has become a critical requirement due to digital transformation, cloud adoption, online payments, and remote work culture. Content beyond syllabus equips students with real-world exposure and prepares them for professional cyber security roles.

Advanced Topics:

Security Operations Center (SOC):

A SOC is a centralized facility responsible for continuous monitoring, detection, analysis, and response to cyber security incidents. Businesses rely on SOC teams to ensure round-the-clock protection.

SIEM (Security Information and Event Management):

SIEM tools collect logs from servers, firewalls, applications, and endpoints. They analyze security events and generate alerts for suspicious activities.

Examples: Splunk, IBM QRadar, ArcSight

Cloud Security:

Cloud security protects data, applications, and infrastructure hosted on cloud platforms like AWS, Azure, and Google Cloud.

Key Concepts:

- Shared Responsibility Model
- Encryption
- Identity and Access Management (IAM)
- Cloud Security Posture Management (CSPM)

Zero Trust Security Model:

Zero Trust follows the principle "Never trust, always verify". Every user and device must be authenticated before access.

Endpoint Detection and Response (EDR):

EDR solutions provide real-time monitoring, threat detection, and automated response on endpoints.

Artificial Intelligence in Cyber Security:

AI is used to detect anomalies, predict attacks, and automate incident response.

Cyber Security Risk Management:

Organizations assess cyber risks using frameworks such as ISO 27001 and NIST.

Real-World Business Case Studies:

Case Study 1: Cloud Data Breach

Misconfigured cloud storage resulted in customer data exposure.



Case Study 2: Ransomware Attack

Business operations were halted due to ransomware infection.

Case Study 3: Phishing Attack

Employee credentials were compromised due to phishing emails.

Learning Outcomes:

- Understanding advanced security concepts
- Awareness of real business cyber threats
- Improved analytical and ethical thinking

Advantages:

- Industry readiness
- Better employability
- Practical business knowledge

Result:

Studying cyber security beyond syllabus prepares students for real-world business challenges and professional cyber security roles.

Viva Questions and Answers

Q1. What is SOC?

A centralized unit for monitoring and responding to cyber threats.

Q2. What is SIEM?

A system that collects and analyzes security logs.

Q3. What is Zero Trust?

A security model that verifies every access request.

Q4. Why is cloud security important?

To protect sensitive business data stored in cloud platforms.

Q5. What is EDR?

Endpoint security solution providing advanced threat detection.